

Younghoo Henry Cho

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Summary

Engineer who builds the infrastructure that makes AI accountable — so organizations can trust what AI tells them and verify why. Seven years across aquaculture production AI, agricultural data infrastructure, and cross-organizational distributed systems, taking ownership from problem definition through production deployment. Two filed patents and peer-reviewed publications.

Technical Skills

Backend & Distributed Systems: Hyperledger Fabric, Go, Node.js, Express, REST APIs, gRPC, microservices

Data & Pipelines: Apache Airflow, Apache Spark, Apache Kafka, data lineage, ETL, PostgreSQL, MongoDB, CouchDB, Firebase

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Git, Linux

Applied AI & Vision: LLM integration, agentic systems, prompt engineering, evaluation frameworks, PyTorch, YOLO, SAM, OpenCV

Languages: Python, Go, JavaScript, Java, C, C++, R

Experience

Graduate Research Assistant, University of Florida

Aug 2022 – Aug 2026

- Architected and shipped DEMETER end-to-end — a platform where independent organizations can share, verify, and trace agricultural data without a central authority — across four independently operated organizations.
- Built a distributed-ledger backend on Hyperledger Fabric (Go) with a Node.js/Express REST API service layer, enabling real-time data lineage tracing across 8,000+ linked records with sub-5s latency.
- Designed scalable event-driven ingestion pipelines using Airflow orchestration, Kafka event streams, and Spark batch aggregations on file-level metadata.
- Integrated a local LLM as a natural language query interface with deterministic routing and a confirmation gate that enforces data correctness before any write operation.
- Designed the platform for generic applicability across agricultural data-sharing scenarios — covering data verification, transformation integrity, and provenance tracing — with production-grade documentation independently adopted by external collaborators without direct handoff.

Researcher, Ministry of Agriculture, Food and Rural Affairs (Korea)

2024 – 2025

- Conducted end-to-end technical evaluation of agricultural mechanization systems across South Korea and the United States, assessing field data requirements and technology adoption gaps to inform cross-country policy decisions.

AI/ML Engineer, Billion21

2019 – 2022

- Built and deployed production computer vision pipelines (YOLO, SAM, OpenCV) for real-time marine monitoring and automated feeding, reducing manual inspection effort in live commercial aquaculture operations.
- Designed and shipped detection and segmentation models from prototype to production, with core components resulting in two filed patents covering blockchain-tracked seaweed farming and spore selection.

Undergraduate Research Assistant, Texas Tech University

2021 – 2022

- Developed computer vision models for biological specimen detection and classification (plant species, animal behavior, acoustic signals), and built data analysis and visualization pipelines to interpret detection outputs across research datasets.

AI Technology Consultant, Global BioAg Linkages

2020 – 2021

- Evaluated AI technology fit for agricultural production systems end-to-end — from technical feasibility assessment and data requirement scoping to market viability analysis — producing deployment recommendations for downstream pilot projects.

Education

University of Florida — Ph.D., Agricultural and Biological Engineering, GPA: 3.95/4.0

Aug 2022 – Expected Aug 2026

Texas Tech University — B.S., Computer Science, Minor in Mathematics, GPA: 3.8/4.0

Aug 2015 – Jul 2022

Selected Publications

- Cho, Y., Yu, Z., & Ampatzidis, Y. *DEMETER: A Blockchain-Based Platform for Tracking the History of Shared Agricultural Data Files*. Under review.
- Cho, Y., Yu, Z., & Ampatzidis, Y. *Facilitating a Future Agricultural Data Ecosystem: A Cross-Sectoral Review of Blockchain Application in Data Sharing*. Under review.
- Cho, Y., Yu, Z., & Ampatzidis, Y. (2025). *Bridging Carbon Markets and Agriculture: A Comparative Study of Stakeholder Engagement in South Korea and United States*. *Discover Agriculture*, 3, 126.

Patents

- *Blockchain-Based Seaweed Farming Data Management and Smart Supply Chain Tracking System*. KR10-2024-0011399, 2024. Status: Pending.
- *Methods for Selecting Spores from Laver Farm Devices*. KR102034354B1, 2019. Status: Granted.